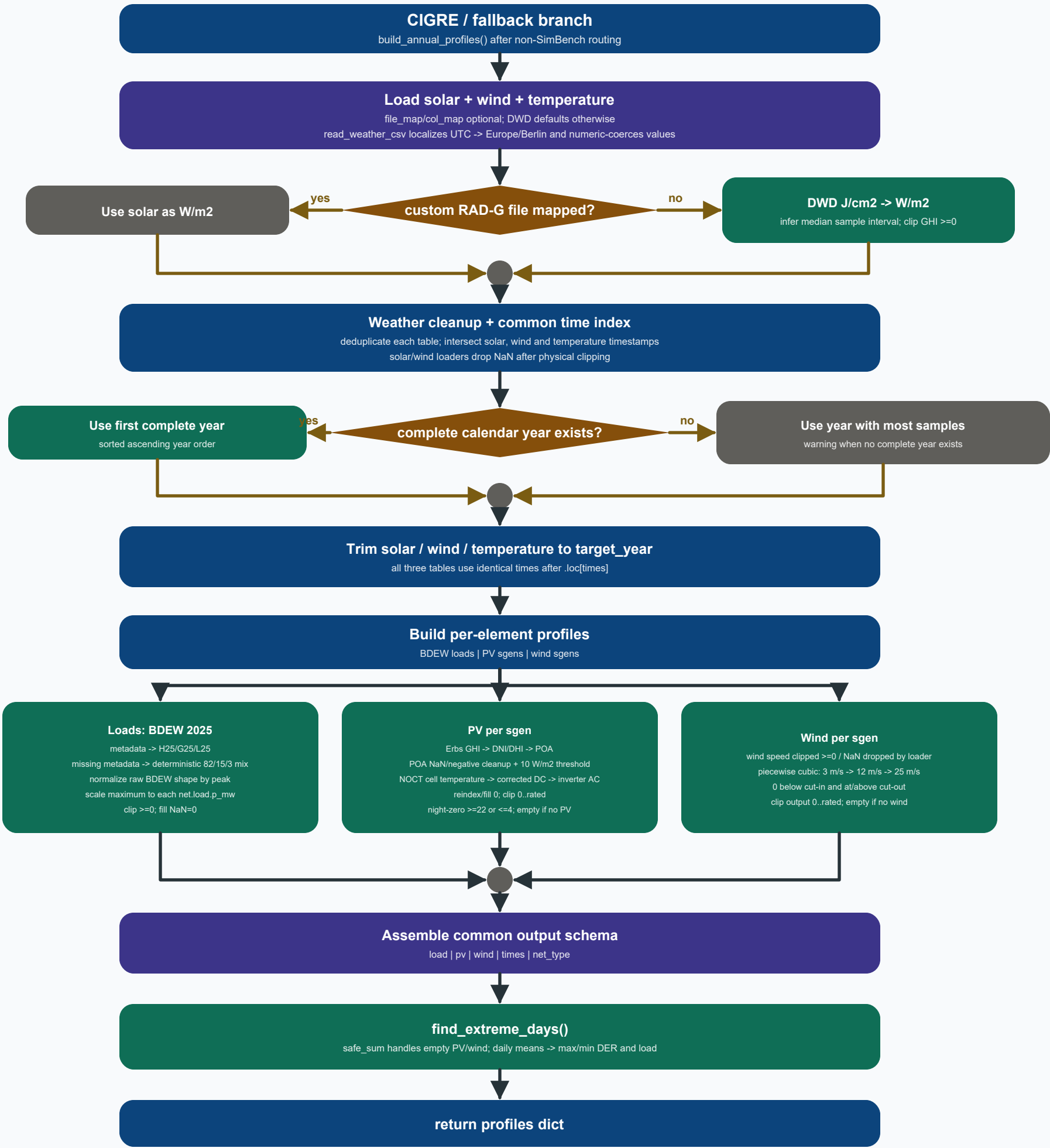




Weather ingestion + cleanup

find_dwd_file(): explicit file_map first; otherwise DWD CDC glob naming.

read_weather_csv(): parse timestamps, sort, UTC-localize, convert to Europe/Berlin.

Values are numeric-coerced; non-numeric source entries become NaN.

Solar: explicit RAD-G mapping is assumed W/m2; otherwise DWD J/cm2 is converted.

DWD solar interval is median timestamp spacing, with 600 s only as short-file fallback.

Solar GHI and wind speed are clipped >=0 and NaNs are dropped by their loaders.

Temperature is assumed degC and NaNs are dropped; then all timestamps are intersected.

Per-element conversion models

Loads: H25/G25/L25 from oemof.demand; semantic metadata is used first.

Unclassified loads receive seeded 82% H25 / 15% G25 / 3% L25 assignment.

BDEW time variation is preserved: raw shape / raw peak, then multiply by load p_mw.

Thus net.load.p_mw anchors the profile maximum, not a constant demand value.

PV: Erbs -> POA -> NOCT temperature -> efficiency correction -> DC -> inverter AC.

PV cleanup: POA NaN/negative cleanup, 10 W/m2 cutoff, rated clipping, night-zero.

Wind: 0 below 3 m/s, cubic to 12 m/s, rated to 25 m/s, then 0 at cut-out.

Wind output is clipped 0..rated; absent PV/wind types leave an empty DataFrame.

Year selection, result schema, and edge cases

Implementation note: the code scans candidate years in ascending order and breaks on the first complete year.

Therefore the current implementation selects the earliest complete year, despite nearby wording that suggests 'most recent'.

If no complete year exists, the code chooses the year with the most aligned timesteps and emits a warning.

All three profile families are returned with the common schema load, pv, wind, times, and net_type.

If no matching PV or wind sgen types are found, the corresponding DataFrame remains empty.

safe_sum in find_extreme_days keeps daily aggregation valid even when PV or wind frames are empty.

The fallback path computes extreme days after assembling the common schema.